

Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; AMV, analytical method validation; ANOVA, analysis of variance; CEX, cation exchange chromatography; Cpk, one-sided process capability index; CPP, critical process parameter; CQA, critical quality attribute; CV, column volume; %CV, coefficient of variation; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSD, relative standard deviation; RSM, response surface methodology; SDM, scale-down model; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

Executive summary

Protein A affinity chromatography is the capture step of the A-Mab drug substance process and the first chromatographic operation in the purification train. It is the step at which leached Protein A enters the process, and it provides the principal clearance of host cell protein and residual DNA. This report presents the characterization executed under PCP-005 on a qualified scale-down model, in support of the transfer of the drug substance process to commercial manufacture. The study follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8, Q9 and Q11, with the published A-Mab case study as the source of the process model (CMC Biotech Working Group 2009).

6 parameters were characterized. 4 were carried into a multivariate design and 2 were assessed one at a time. The outcome is 2 well-controlled critical process parameters, 2 key process parameters and 2 general process parameters. No parameter at this step is a critical process parameter, which is consistent with the platform experience described in Section 2.

The multivariate work comprised a screening design of 19 runs and a face-centred central composite design of 28 runs in protein load, elution buffer pH, load flow rate and end of pool collect. The screening design identified which factors are active. The response-surface model is the predictive model and the basis of the design space. Pool HCP is described by a strong surface ($R^2 = 0.989$, predicted $R^2 = 0.932$), on which elution buffer pH carries the largest main effect. The step yield model is adequate. No factor effect on leached Protein A was resolved, and no predictive claim is made for that response.

Leached Protein A is the one CQA this step sets. The highest value measured anywhere in the two designs was 4.61 ppm, against an acceptance criterion of 0–5 ppm, and the fitted model gives 2.87 ppm at the set-point and a mean of no more than 4.35 ppm across the characterized region. That model resolves no factor effect, so the measured figure is the one that bounds the result. Once the pure error of the response is propagated, the upper bound of the 95 % predictive interval reaches the criterion above elution buffer pH 3.86, which is why the design space is capped there (Section 7). At the drug substance the attribute carries a capability of $C_{pk} = 2,785$, which is very large because the polishing steps clear what capture introduces. Of the three attributes this step governs, host cell protein has the tightest capability at $C_{pk} = 6.14$, with a mean of 15.2 ng/mg against a limit of 100 ng/mg.

2 deviations were recorded during execution. Both were investigated, impact-assessed and dispositioned as retained, and neither altered a parameter classification or the operating region defined in Section 6. They are described in Section 13.

No claim is made that this step assures the drug substance limit for host cell protein. The capture pool carries 72 times that limit at the set-point, and even the lowest value anywhere in the characterized region is 15 times it, so the cation and anion exchange steps deliver the remaining clearance (PCR-007 and PCR-008). No viral clearance claim is made for the Protein A step. Within those bounds the step is well understood over the ranges studied, its operating region is defined, and its results roll up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cellular cytotoxicity. It is produced by fed-batch cell culture and purified by a platform train of six operations. Protein A capture is the first of those operations, and it is followed by low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration. The train and the role of each step are described in PTP-001.

The step uses an immobilized Protein A resin operated in bind and elute mode. Clarified harvest is loaded onto the equilibrated column, process impurities (host cell protein, DNA and small molecules) pass into the flow through or are removed in the wash, and the antibody is eluted with a low pH buffer. The elution pool is the feed to the low-pH hold, so the elution buffer pH set here also conditions the following step. Capture therefore does three things at once: it recovers the product, it removes the bulk of the impurity burden carried out of cell culture, and it introduces a small amount of ligand released from the resin.

That released ligand is the quality attribute the step sets. Leached Protein A is a process impurity of low to moderate criticality with an acceptance criterion of 0–5 ppm (Table 2.1). Host cell protein (HCP) and residual DNA are formed upstream and are reduced here, not set here. The nominal batch leaves capture at 9,125 ng/mg of host cell protein, and residual DNA is reduced by 2.5 log₁₀ across the step, so neither attribute meets its drug substance criterion at this point in the train and neither is expected to. The cross-step position is set out in Section 10.

1.2 Objectives

The objectives of the study were:

- (i) to quantify the relationship between the operating parameters of the capture step and the responses that describe its product quality and process performance;
- (ii) to define an operating region over which leached Protein A remains within its acceptance criterion and step yield remains acceptable;
- (iii) to establish proven acceptable ranges for each characterized parameter and to confirm that the normal operating ranges sit inside them;
- (iv) to classify each parameter against the criteria of SOP-4001; and
- (v) to justify the ranges and controls carried into Stage 2 and into the control strategy for the drug substance.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). Its content maps to the record that stage expects: the sources of variability were identified, their effect on the responses was measured, and the resulting understanding was converted into a control strategy. The enhanced development approach of ICH Q8 provides the design-space framework, ICH Q11 the development and manufacturing-process content, and ICH Q9 the risk methodology, under which criticality is treated as a continuum and not as a binary label, following ICH Q9 itself and the A-Mab case study (CMC Biotech Working Group 2009).

No viral clearance is claimed for this step, so ICH Q5A is not applied here. Viral clearance is claimed for low-pH inactivation, anion exchange and small-virus retentive filtration, and is consolidated in PCMR-001. The paired protocol for this work is PCP-005, which fixed the design, the acceptance criteria and the analysis before execution.

1.4 Related documents

The documents that frame this report are listed in Table 1.1, and the controlled procedures and validated methods it relies on are listed in Table 1.2.

Table 1.1: Documents related to this report.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-005	Process Characterization Plan (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 1.2: Controlled procedures and validated analytical methods for the Protein A step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP

Reference	Title	Type
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Protein A capture is a well established operation in the Novacyte purification platform. The same resin, the same buffer system and the same mode of operation have been used to manufacture three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and to supply the A-Mab clinical programme, so the behaviour of the step is established from a large body of performance history rather than from this study alone. Affinity capture is inherently robust, because the separation rests on a specific interaction between the ligand and the Fc region and not on a balance of weak forces that shifts with buffer composition. The purpose of this study was therefore to confirm and bound known platform behaviour, not to discover it.

The platform sets a clear mechanistic expectation for each parameter, and that expectation framed the ranges. Pool HCP is expected to rise as protein load increases, because the mass of co-adsorbed impurity carried into the elution pool grows with the mass of product bound. Pool HCP is also expected to rise as elution buffer pH falls, because a more aggressive elution desorbs species that are held less tightly than the antibody. Those two effects are expected to interact, since the impurity available for desorption depends on how much material was loaded. Step yield is expected to fall when a high load is combined with a high linear velocity, because the contact time then becomes short relative to the rate of binding. End of pool collect is expected to trade yield against pool volume and purity.

Leached Protein A is treated differently. Ligand release is a property of the resin, its chemistry and its cycle history, and the low pH of the elution is the driver in principle. The platform expectation was that release is small and roughly constant across the operating window, and that its cycle-to-cycle variability is comparable to the precision of the assay. This study was designed to test that expectation and not to assume it.

Prior knowledge is bounded in one respect that matters for the transfer. The platform evidence covers the resin used in this characterization. An alternate Protein A resin was assessed separately in RA-004 (2025-11-03), which concluded that independent characterization would be required and that bridging from platform data would not be acceptable. Nothing in this report supports a resin change.

2.2 Quality attributes in scope

The attribute this step sets is given in Table 2.1, and the attributes carried into the step from upstream are given in Table 2.2. Criticality follows the A-Mab Tool #1

assessment, in which the impact of an attribute is combined with the uncertainty attached to it, and attributes scoring high or very high are called critical (CMC Biotech Working Group 2009).

Table 2.1: The quality attribute set by the Protein A step, with its criticality from the A-Mab Tool #1 assessment.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0-5 ppm	L-M	16

Table 2.2: Attributes carried into the Protein A step from upstream, which the step reduces or passes through.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60

Leached Protein A is the only entry in Table 2.1, and its criticality is low to moderate. It is nonetheless carried in the register as a critical quality attribute, because the register scores criticality as a continuum and does not drop an attribute that falls below the threshold for the label. Two features of the attribute drive the study design. It is introduced here and nowhere else, so no upstream control acts on it. It is also cleared by the two ion exchange steps, so the level in the drug substance is not the level in the capture pool.

Of the attributes in Table 2.2, host cell protein is the one this step influences most, and it is the response the multivariate design was built around. Residual DNA is of very low criticality and is reduced by a fixed mechanism, since DNA does not bind the ligand and passes into the flow through. Aggregates carry the highest criticality of the three in Table 2.2, but they are formed in the bioreactor and are not resolved by affinity capture, so the step is treated as passing them through unchanged (their polish is the subject of PCR-007).

2.3 Risk-based prioritization and parameter selection

RA-001 set the scope of this study before any of it was executed. The method there is risk ranking and filtering: each parameter is scored for its potential impact on a quality attribute and separately for its potential to interact with other parameters, and the combined score decides the type of study it receives (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b). Parameters with no

plausible link to an attribute and a low interaction score are not studied and take their ranges from the knowledge space (there are none of these at the capture step).

Four parameters were assigned to the multivariate design. Protein load and elution buffer pH scored on the quality axis because both act on pool HCP, and both scored on the interaction axis because the impurity mass available for desorption depends on the load. Load flow rate and end of pool collect scored on process performance and not on product quality, and were carried into the same design because the yield effect of flow depends on the load. Assigning all four to one design is the reason the study can separate a load effect from a load-by-flow effect.

Operating temperature and bed height were assigned to univariate assessment. Neither has a mechanism by which it could change the chemistry of the separation over the range of interest, since bed height acts through residence time (which is already varied through load and flow) and operating temperature acts on binding kinetics only weakly over the range the facility can produce. Both were nonetheless assessed across their characterization ranges, because a parameter that is not studied cannot be given a proven acceptable range.

The ranges themselves were set at about twice the routine control range for each parameter, following the platform convention for characterization studies. The wider ranges do two things. They allow the study to see whether an effect that is invisible inside the NOR appears at a distance from it, and they leave room to move the operating point later without a new study.

3 Materials and methods

3.1 Scale-down model and its qualification

The characterization was executed on a qualified laboratory-scale model of the commercial capture step, operated under SOP-2007 and qualified under SOP-1001. The model was designed on the established principles of chromatography scaling: bed height is held at the commercial value of 20 cm, linear velocity is matched in cm/hr, protein load is matched in g of product per L of resin, and the equilibration, wash and elution volumes are matched as column volumes. Column efficiency was verified by plate count and peak asymmetry before each execution.

Qualification compared the small-scale system against commercial-equivalent runs on the inputs and the outputs that matter to this step (step yield, pool volume, pool host cell protein, residual DNA and leached Protein A), and no statistically significant difference in column efficiency or in any performance measure was found between the two scales. The qualification data are held in the SOP-1001 qualification record for the capture model and are not reproduced here. On that basis the model is treated as representative of the commercial step, and the operating ranges and clearance claimed below are claimed for the commercial step through it.

Two features of the model bound what it can support. It reproduces the hydrodynamics and the chemistry of the separation, but it does not reproduce the resin cycle history of a commercial campaign, so all runs were executed on resin within the early part of its validated life. It also uses a single representative clarified harvest as the feed, so the study varies this step's parameters and not the properties of the material entering it. The consequence of both bounds is stated in Section 11.

The runs were executed in the temperature-controlled chromatography chamber EQ-CHR-118, which was in calibration at execution and whose next calibration was due 2026-05-30.

3.2 Operation

Each run followed the commercial sequence. The column was equilibrated, loaded to the target protein load at the target linear velocity, washed, and eluted with the low pH elution buffer, and collection started on the ascending edge of the elution peak and stopped at the target end of pool collect, expressed in column volumes. The pool was sampled, then titrated and held for the low-pH inactivation step. Set-points and ranges are given in Table 4.1.

Buffers were prepared and released under SOP-2103 (Buffer Preparation, Release and Expiry Control), and each elution buffer lot was released on measured pH before use (see Section 13). Resin packing, sanitization and cycle accounting follow SOP-2008. Resin cycle number was held constant across the study and was not a factor of the design.

3.3 Analytical methods

Each of the two impurity responses is measured by one validated method. Pool host cell protein is measured by the ELISA validated under AMV-3012. Leached Protein A is measured by the ELISA validated under AMV-3016. Size variants are measured by SEC-HPLC (AMV-3011) and residual DNA by qPCR (AMV-3014). Both support the in-process panel and neither is a response of the multivariate design. Step yield is determined by protein mass balance across the step under SOP-2007. The validated performance of the two ELISA methods is given in Table 14.12 in Appendix D.

Two points about method performance carry into the interpretation of the results. The leached Protein A ELISA has an intermediate precision of 6.5 %RSD and a limit of quantitation of 0.25 ppm, so results in the low single digits are quantitative but not sharp, and the host cell protein ELISA has an intermediate precision of 9.5 %RSD. Where a centre-point spread reported below is larger than these figures, the excess is run-to-run process variation and is treated as such. Where it is smaller, the spread lies within the validated precision of the method, and no process contribution can be separated from assay noise on the replicates available.

3.4 Sampling plan

One elution pool sample was taken per run and assayed for host cell protein and leached Protein A. Load and flow-through samples were taken on every run to close the mass balance and to confirm the input to the step. Replication is carried by the centre points rather than by repeat assays: the screening design carries 3 centre-point runs and the response-surface design carries 4 (Appendix A and Appendix B). Those replicates are independent executions at the set-point, so their spread contains both the process and the assay contribution, and they provide the pure-error term for the lack-of-fit tests in Section 5.

3.5 Statistical methods

Factors are analysed in coded units, in which the set-point is 0 and the edges of the characterization range are the two coded extremes (−1 and +1). Coding makes the effect estimates comparable across parameters with different units, and it means an effect is read as the change in the response between the centre of the range and its edge.

The screening design is analysed as an ordinary least-squares model in the main effects and the two-factor interactions. An effect is reported as twice the coded coefficient, so that it is the change across the whole characterization range. The response-surface design is analysed as a full quadratic model in the same coded factors. Significance is assessed at $\alpha = 0.05$ throughout, and no term is removed from a reported model on the basis of its p-value.

Model adequacy is judged on four things together: the coefficient of determination with its adjusted and predicted values, the overall F test, and an ANOVA partition of the residual into lack of fit and pure error (tested against the centre-point replicates). The predicted value is computed by leave-one-out prediction, so a model that fits its own data and predicts a held-out run poorly is visible. A response for which no term is significant is reported as robust to the factors over the ranges studied, and the factors are retained in the knowledge space as evidence of that robustness.

Proven acceptable ranges are computed from the fitted response-surface model in two ways, as PCP-005 specifies. The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range. The second propagates the normal operating ranges of the other parameters by Monte-Carlo of the fitted model, and keeps the part of the range over which the upper bound of the 95 % predictive interval still meets the criterion. Commercial-scale capability is estimated by Monte-Carlo of 2,000 batches with every parameter varying inside its NOR, and is reported as a one-sided Cpk against the drug substance criterion.

4 Study design

4.1 Factors, ranges and the knowledge space

The design resolves the active factors and then maps the region they define. Table 4.1 gives every parameter with its set-point, its normal operating range, the range studied and the classification the study produced. The coded letters used in the effect and coefficient tables are given in Table 4.2.

Table 4.1: Parameters of the Protein A step, with the ranges studied and the classification resulting from this study.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	30	20-40	10-50	WC- CPP	multivari- ate
Elution buffer pH	pH	3.55	3.4-3.7	3.2-3.9	WC- CPP	multivari- ate
Load flow rate	cm/hr	200	150-250	100-300	KPP	multivari- ate
End of pool collect	CV	2.6	2.3-2.9	2-3.2	KPP	multivari- ate
Operating temperature	°C	22	18-26	15-30	GPP	univariate
Bed height	cm	20	15-25	10-30	GPP	univariate

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

The characterization range in Table 4.1 is the knowledge space of this step, not a proven acceptable range. Each range is about twice the corresponding NOR, and each was chosen against a mechanism. Protein load spans 10 to 50 g/L resin, which reaches well past the point at which the platform expects the impurity carried into the pool to grow, while staying below the static binding capacity of the resin. Elution

buffer pH spans 3.2 to 3.9. The low edge was set above the pH at which antibody precipitation becomes a risk during elution, and the high edge was set at the point where elution becomes incomplete. Load flow rate spans 100 to 300 cm/hr, which covers the range of linear velocities the commercial skid can deliver at this bed height, and end of pool collect spans 2 to 3.2 CV, from a cut that sacrifices yield to one that collects deep into the descending edge.

4.2 Screening design

The screening study was a two-level full factorial in the four multivariate parameters, augmented with centre points. It comprised 19 runs in total: 16 factorial runs and 3 replicated centre points. The design estimates all four main effects and all six two-factor interactions without aliasing, and the centre points give an independent estimate of pure error and a test for curvature (a two-level design cannot separate curvature from the mean without them). The full design matrix with the measured responses is given in Appendix A.

The purpose of this design is to identify which factors have an important statistical effect on each response. It is not the predictive model. A two-level design carries no information about curvature within the range it spans, so the magnitude of an effect it estimates is refined by the response-surface study that follows.

4.3 Response-surface design

All four screening factors were carried into the response-surface study, because the screening results left none of them without an effect on at least one response. The design is a face-centred central composite design of 28 runs: 16 factorial runs, 8 axial runs on the faces of the cube and 4 replicated centre points. Placing the axial points on the faces keeps every run inside the characterization ranges, which matters here because those ranges are also the edges of the knowledge space. The design supports the full quadratic model of 15 terms and leaves residual degrees of freedom for the lack-of-fit test. The matrix and the responses are given in Appendix B.

This is the predictive model of the step and the basis of the design space in Section 6.

4.4 Univariate assessment

Operating temperature and bed height were assessed one parameter at a time across the ranges given in Table 4.1, with the remaining parameters at their set-points. Pool host cell protein, leached Protein A and step yield were measured on each run (the same responses as the multivariate design). Neither parameter was linked to a quality attribute, and both were classified as general process parameters on that

basis (Section 9). The supporting data are held in the study data package referenced by PCP-005.

No response surface was fitted for either parameter, so no multivariate claim is made for them. Their acceptable ranges are the ranges assessed, and they are not part of the region defined in Section 6.

5 Results

5.1 Centre-point performance and reproducibility

Reproducibility at the set-point differs sharply between the three responses, and it sets what each of the models below can be expected to resolve. Table 5.1 gives the centre-point mean, standard deviation and coefficient of variation for the response-surface design, and Table 5.2 gives the same for the screening design.

Table 5.1: Centre-point reproducibility in the response-surface design, from 4 replicated runs at the set-point.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	7.56e+03	357	4.72
Step yield	4	0.946	0.0187	1.97
Leached Protein A (ppm)	4	2.86	0.967	33.8

Table 5.2: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	3	8.1e+03	1.61e+03	19.9
Step yield	3	0.935	0.00599	0.64
Leached Protein A (ppm)	3	3.28	0.508	15.5

Step yield is the most reproducible response, at 2.0 % in the response-surface design. Pool HCP reproduces to 4.7 % there, which is well inside the effects reported below. Leached Protein A is the least reproducible response, at 33.8 % across the response-surface centre points and 15.5 % across the screening centre points, and that spread is several times the intermediate precision of AMV-3016, so most of it is run-to-run variation in the amount of ligand released and not assay noise. A response with this much pure error can only resolve a large effect, which is the correct context for the leached Protein A results below.

Pool HCP behaves differently in the two designs. Its screening centre points give 19.9 % against 4.7 % in the response-surface design, on 3 and 4 replicates. The screening figure is above the 9.5 %RSD intermediate precision validated for AMV-3012 and the response-surface figure is well below it, so the two designs give different accounts of how reproducible this response is. Each design supplies the pure error for its own

lack-of-fit test, so the wider screening spread does not enter the response-surface tests in Section 5.3.

The centre points also give the pure-error term used in every lack-of-fit test in this section. They are independent executions at the set-point and not repeat injections of one sample, so their spread is intended to carry the process contribution as well as the assay contribution. For pool HCP in the response-surface design that cannot be shown, because the spread is smaller than the validated precision of the method that measured it. A pure-error estimate on 3 degrees of freedom is imprecise in both directions, and the lack-of-fit result for pool HCP below passes by a narrow margin, so the two should be read together.

5.2 Screening: factor effects

The screening design separated the active factors for pool HCP and for step yield, and found none for leached Protein A. Table 5.3 gives the model summary for the three responses.

Table 5.3: Screening model summary. The model for leached Protein A is not significant overall.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.973	0.939	0.886	28.6	3.49e-05	1.4e+03
Step yield	19	0.976	0.945	0.798	32	2.28e-05	0.00833
Leached Protein A (ppm)	19	0.371	-0.414	-2.98	0.473	0.867	0.513

The two significant models in Table 5.3 behave very differently from the third (leached Protein A). Pool HCP and step yield are both fitted with R² above 0.97 and overall p-values below 0.05. The leached Protein A model has an overall p-value of 0.87 and a negative adjusted R², which is what a model returns when its terms explain less than the degrees of freedom they consume, and no factor effect on leached Protein A was resolved by the screening design.

Pool HCP is driven by elution buffer pH and protein load, and by the interaction between them (Table 5.4). Lowering elution buffer pH across the range raises pool HCP by 8,840 ng/mg, which is the largest effect estimated for this response. Raising protein load across its range raises pool HCP by 7,269 ng/mg. The load by elution pH interaction is significant and negative, so the penalty from a high load is smaller when the elution is milder. Neither load flow rate nor end of pool collect reached significance for this response ($p > 0.05$ for both).

Table 5.4: Largest screening effects on pool HCP. The full table is in Appendix C.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	

Step yield is driven by end of pool collect, and then by protein load and load flow rate (Table 5.5). Collecting further into the descending edge of the elution peak raises yield by 6.0% across the range, which is the dominant term. Load and flow each reduce yield, and their interaction is significant, so the two act together and not separately. The remaining interactions that reach significance are small and are reported for completeness in Appendix C.

Table 5.5: Largest screening effects on step yield. The full table is in Appendix C.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*

These are identifications, not predictions. The screening design places every run at an edge or at the centre, so it can say which factors matter and in which direction, but it cannot describe how a response bends between those points. The magnitudes above are refined by the response-surface models in Section 5.3, and the design space rests on those models and not on this one.

5.3 Response-surface models

The response-surface study confirms the screening result for pool HCP and for step yield, adds the curvature the two-level design could not see, and again resolves nothing for leached Protein A. Table 5.6 gives the model summary.

Table 5.6: Response-surface model summary for the three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.989	0.976	0.932	80.9	2.71e-10	895
Step yield	28	0.913	0.818	0.586	9.69	0.000105	0.016

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Leached Protein A (ppm)	28	0.567	0.101	-0.714	1.22	0.365	0.637

Only one of these three models is predictive. Pool HCP is fitted with $R^2 = 0.989$, adjusted $R^2 = 0.976$ and predicted $R^2 = 0.932$, and the gap between the adjusted and predicted values is small, so the model predicts a held-out run about as well as it fits the runs it was built on. Step yield is fitted with $R^2 = 0.913$ and an overall p-value of $1.0\text{e-}04$, but its predicted R^2 of 0.586 is much lower than its adjusted value. That model is used below to describe the direction and the size of the yield effects and to bound the yield across the region. Its predicted R^2 is the limit on how closely any single value read off it should be taken, and the yield figures quoted in Section 6 and Section 8 carry that limit with them. The leached Protein A model is not significant overall ($p = 0.36$) and its predicted R^2 is negative, so no predictive claim is made for it at all.

The coefficient table for pool HCP is given in Table 5.7, and the ANOVA partition with the lack-of-fit test is given in Table 5.8.

Table 5.7: Full quadratic coefficients for pool HCP, in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7,215.5	309	23.3	5.44e-12	***
A	3,871.8	211	18.4	1.12e-10	***
B	-5,244.6	211	-24.9	2.4e-12	***
C	-435.4	211	-2.06	0.0596	
D	-1,055.5	211	-5	0.000242	***
AB	-1,899.8	224	-8.49	1.16e-06	***
AC	-429.1	224	-1.92	0.0774	
AD	-518.1	224	-2.32	0.0376	*
BC	348.2	224	1.56	0.144	
BD	727.4	224	3.25	0.00632	**
CD	-128.7	224	-0.575	0.575	
A ²	-344.5	557	-0.618	0.547	
B ²	1,943.5	557	3.49	0.00401	**
C ²	851.3	557	1.53	0.151	
D ²	330.5	557	0.593	0.563	

Table 5.8: ANOVA partition for the pool HCP model, with lack of fit tested against the centre-point pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05	nan	nan
Lack of fit	1e+07	10	1e+06	7.88	0.058

Source	Sum sq.	df	Mean sq.	F	p-value
Pure error	3.82e+05	3	1.27e+05	nan	nan

Four terms of Table 5.7 carry most of the surface. Elution buffer pH has the largest coefficient, protein load the second largest, and the interaction between them is significant and negative as it was in the screening study. The quadratic term in elution buffer pH is significant and positive, which is the curvature the screening design could not detect: pool HCP falls steeply as pH rises from the low edge and then flattens. End of pool collect (term D) also carries a significant negative coefficient, at 20.1% of the elution pH coefficient in magnitude. Its effect on this response is real and it is small.

Two smaller interactions also reach 0.05. Protein load with end of pool collect and elution buffer pH with end of pool collect carry 27.3% and 38.3% of the load by elution pH coefficient in magnitude. Both are reported in Table 5.7 and neither is interpreted further here. Both involve end of pool collect, which is held at its set-point in the surfaces shown below, so neither contributes to those panels.

The lack-of-fit test in Table 5.8 does not reject the model ($p = 0.058$), but the margin above 0.05 is narrow. The pure-error term comes from only 4 centre points, so the test has little power, and a narrow pass should not be read as a demonstration that the quadratic form is complete. The diagnostics in Figure 5.1 support the model on the other axes. Standardized residuals show no structure against the predicted value, the normal quantile plot is close to linear, and the actual against predicted plot sits on the identity line across the whole range of the response.

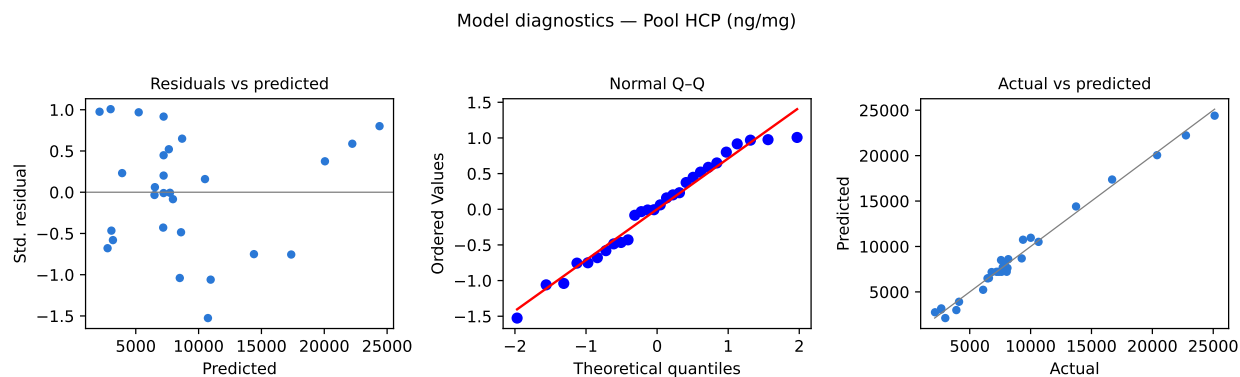


Figure 5.1: Model diagnostics for the pool HCP response surface: standardized residuals against predicted value, normal quantile plot, and actual against predicted.

The fitted surfaces are shown in Figure 5.2 over the plane of protein load and elution buffer pH, with the other two parameters at their set-points, and the pool HCP panel shows the diagonal structure the interaction implies: the contours are steep and closely spaced at low elution pH, and they spread out as pH rises. At the low pH edge of the panel the fitted model moves from 8,288 to 19,831 ng/mg between the lowest

and the highest load. At the high pH edge the same change in load moves it from 1,598 to 5,542 ng/mg, so the penalty a high load carries is 2.9 times larger in ng/mg at the low pH edge than at the high one. The step yield panel is almost flat in this plane, which is correct, since yield is governed by end of pool collect and by the load with flow interaction (neither of which is an axis of this figure). The leached Protein A panel is shown for completeness. The model behind it is not significant overall, and the one term in it that reaches 0.05 is discussed below.

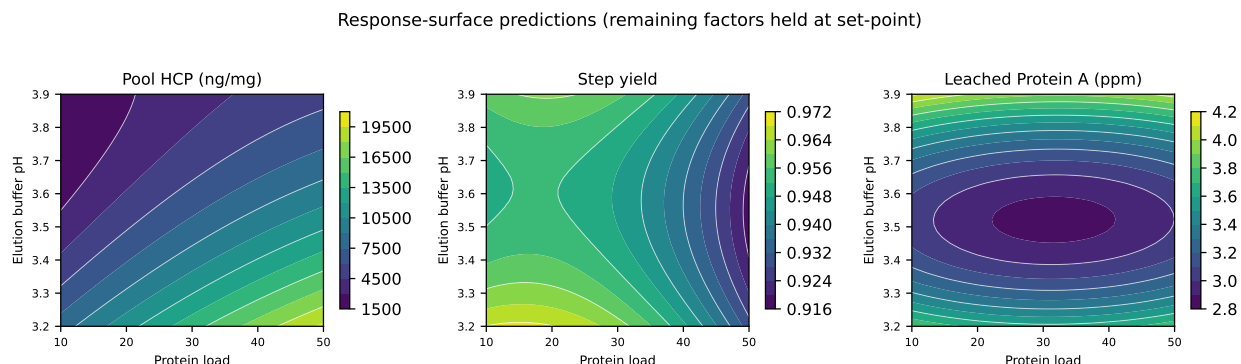


Figure 5.2: Response-surface predictions over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points.

The coefficient tables for step yield and for leached Protein A are given in Appendix C. For step yield the pattern of the screening study is reproduced: end of pool collect carries the largest coefficient, protein load and load flow rate carry smaller negative coefficients, and the load by flow interaction remains significant. For leached Protein A the only term reaching 0.05 is the quadratic in elution buffer pH. Since the model containing it is not significant overall, and since 14 terms were tested in it, that single term is not treated as a demonstrated effect. It is carried into the proven acceptable range analysis in Section 7 because doing so is conservative, and it is identified there as the reason one range is bounded.

5.4 Mechanistic interpretation

The shape of the pool HCP surface follows from how the two mechanisms of impurity carryover combine. Host cell protein reaches the elution pool by two routes: it is co-adsorbed onto the resin and the antibody during the load, and it is desorbed with the antibody when the pH drops. Protein load controls the first route (co-adsorption), because more product bound means more associated impurity retained through the wash. Elution buffer pH controls the second, because a lower pH breaks weaker interactions as well as the specific one. The interaction term is negative because the second route gates the first: impurity that was co-adsorbed only reaches the pool if the elution is aggressive enough to release it.

The positive curvature in elution buffer pH has the same explanation. Between the low edge of the range and about the set-point, each further drop in pH releases another

population of weakly held species, so the response is steep. Above the set-point the elution is already selective and the remaining species stay on the column, so the response flattens. This reproduces the behaviour reported for the capture step in the published case study (CMC Biotech Working Group 2009).

End of pool collect acts on the same quantity from the other side. Pool host cell protein is expressed per mg of product, so a later cut that adds product mass to the denominator lowers the ratio, and the coefficient for that term in Table 5.7 is negative. The platform expectation recorded in Section 2 was that a later cut trades purity for yield. Over the ranges studied the data do not support that for this response, and the sign of the effect is the opposite one.

Step yield is governed by two unrelated mechanisms and the surface shows both. End of pool collect is a simple integration effect, because product continues to elute on the descending edge and a later cut recovers more of it (at the cost of a larger pool). The load by flow interaction is a mass transfer effect. At a high load the binding front reaches further down the bed, and at a high linear velocity the contact time in that front is shorter, so product breaks through into the flow through before it can bind. Neither mechanism touches product quality, which is why both parameters classify on process consistency in Section 9.

Leached Protein A behaved as the platform predicted. Ligand release is set by the chemistry of the resin and by its cycle history, both of which were held constant, and the elution pH range studied here is not wide enough to move it against the run-to-run variability of the step. The null result is informative in two ways. It means the amount of ligand entering the process does not have to be managed by choosing an operating point inside this region, and it means the four factors stay in the knowledge space as evidence that the attribute is robust to them across the ranges studied.

Taken together, these mechanisms are why the operating region has to be multivariate and not a set of four independent ranges. 3 of the significant terms in Table 5.7 are interactions, so the acceptable range of protein load is not a fixed interval. It depends on where elution buffer pH sits. A set of one-dimensional ranges cannot express that, and the design space in Section 6 is stated as a region for exactly this reason.

6 Design space

The design space of the capture step is the characterized region in the four multivariate parameters, bounded above in elution buffer pH by the robustness limit derived in Section 7. Inside that region the attribute this step sets stays within its acceptance criterion when the other parameters move anywhere within their normal operating ranges, and step yield stays above the value given below. The region is defined on the fitted response-surface models of Section 5.3 and not on the screening study.

Two planes carry the structure of the region. The plane of protein load against elution buffer pH is the plane in which pool HCP moves, and it is the plane shown in Figure 5.2. The plane of protein load against load flow rate is the plane in which step yield moves. The two planes are almost independent of each other, since elution buffer pH has no significant effect on yield and load flow rate has none on pool HCP.

The corners and the region extremes below are computed over the full characterization ranges of Table 4.1, which is the wider of the two regions and the conservative one to quote. The design space itself is that region with elution buffer pH capped at 3.86, and capping it lowers the highest predicted leached Protein A from 4.35 to 4.14 ppm. The lowest predicted step yield is the same figure either way, because elution buffer pH carries no significant effect on yield.

The worst case corner for pool HCP is protein load 50 g/L resin, elution buffer pH 3.2, load flow rate 100 cm/hr and end of pool collect 2 CV, with all four parameters at an edge of their characterization ranges. There the fitted model predicts 24,397 ng/mg against 7,216 ng/mg at the set-point and 2,119 ng/mg at the lowest corner, so the step spans more than an order of magnitude in pool HCP across the region it was characterized over. That corner is not excluded from the design space, because pool HCP is not an acceptance criterion at this step. What follows from it is a requirement on the downstream steps, and that is stated in Section 10.

The worst case corner for leached Protein A is protein load 10 g/L resin, elution buffer pH 3.9, load flow rate 300 cm/hr and end of pool collect 2 CV, at which the model predicts 3.99 ppm. The highest mean anywhere in the region, corners and interior together, is 4.35 ppm against an acceptance criterion of 5 ppm. Those are mean levels from a model that resolves no factor effect, so the measured results bound the attribute better than the model does. The highest leached Protein A measured in either design is 4.61 ppm, on response-surface run 7, which is that same corner (Table 14.4).

The worst case corner for step yield is protein load 50 g/L resin, elution buffer pH 3.9, load flow rate 300 cm/hr and end of pool collect 2 CV, at which the model predicts 85.6% against 94.7% at the set-point, and the lowest value anywhere in the region is 84.4%.

The normal operating ranges in Table 4.1 sit inside this region with margin on every parameter. Propagating them through the fitted models gives pool HCP between 3,695 and 13,553 ng/mg, leached Protein A at or below 3.16 ppm, and step yield at or above 90.8%. The nesting is the usual one. The characterization ranges are the knowledge space, the region defined here is the design space, and the normal operating ranges are the control space the process is actually run in.

Movement within the design space is not a change in the sense of ICH Q8. It is still subject to the site change-management system under ICH Q10, and a planned move is assessed prospectively against the risks of that particular move. Movement outside the design space requires change control and, where the move is outside the characterization ranges, new data.

Three bounds apply to this claim. The region is defined only over the ranges in Table 4.1; nothing here supports a setting outside them. The models predict mean levels, so the assurance at the edge of the region is lower than at its centre, and the capability figures in Section 8 are the statement of assurance inside the NOR and not at the edge. The region is defined on a qualified scale-down model with resin early in its validated life and a single representative feed, so it is claimed for the commercial step through the qualification described in Section 3 and no further.

7 Proven acceptable ranges

The acceptance basis has to be stated before the ranges, because it differs between the two governed attributes. Both criteria in this section are the study-provided drug substance specifications: 0–5 ppm for leached Protein A and 0–100 ng/mg for host cell protein. Neither is an in-process limit for the capture pool, and no viral clearance response arises at this step, so no step-level criterion is back-calculated anywhere in this report. Step yield has no acceptance criterion of this kind and is reported as process performance.

Table 7.1 gives, for each governed attribute and each multivariate parameter, the characterization range and both proven acceptable ranges: the range proven with the other parameters at their set-points, and the range proven with the other parameters varying inside their normal operating ranges.

Table 7.1: Proven acceptable ranges by attribute and parameter, from the fitted response-surface models.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Protein load	10–50	g/L resin	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Elution buffer pH	3.2–3.9	pH	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	Load flow rate	100–300	cm/hr	none (set-point breaches)	none (set-point breaches)
Pool HCP (ng/mg)	End of pool collect	2–3.2	CV	none (set-point breaches)	none (set-point breaches)
Leached Protein A (ppm)	Protein load	10–50	g/L resin	10–50	10–50
Leached Protein A (ppm)	Elution buffer pH	3.2–3.9	pH	3.2–3.9	3.2–3.86
Leached Protein A (ppm)	Load flow rate	100–300	cm/hr	100–300	100–300
Leached Protein A (ppm)	End of pool collect	2–3.2	CV	2–3.2	2–3.2

For leached Protein A the whole characterization range of protein load, load flow rate and end of pool collect is proven acceptable under both analyses, which means each of those ranges stays acceptable when the other three parameters move anywhere inside their normal operating ranges. Elution buffer pH is the one exception. Its at set-point range is the full 3.2–3.9, and the NOR-propagated range stops at 3.86. The binding attribute is leached Protein A and the binding statistic is the upper bound of the 95 % predictive interval, which reaches the acceptance criterion near the top of the pH range once the pure error of the response is propagated with it. The NOR upper edge for this parameter is 3.7, so the routine operating range sits inside the proven range with margin.

That bound should be read with caution. It comes from the quadratic term discussed in Section 5.3, in a model that is not significant overall, combined with a pure error of 33.8 % at the centre point. It is therefore a conservative limit produced by an imprecise response, and not a demonstrated edge of failure. Figure 7.1 shows the analysis for the governing parameter of this attribute, which is protein load. The acceptable region is shaded, and the response sits below the criterion across the whole range in both panels.

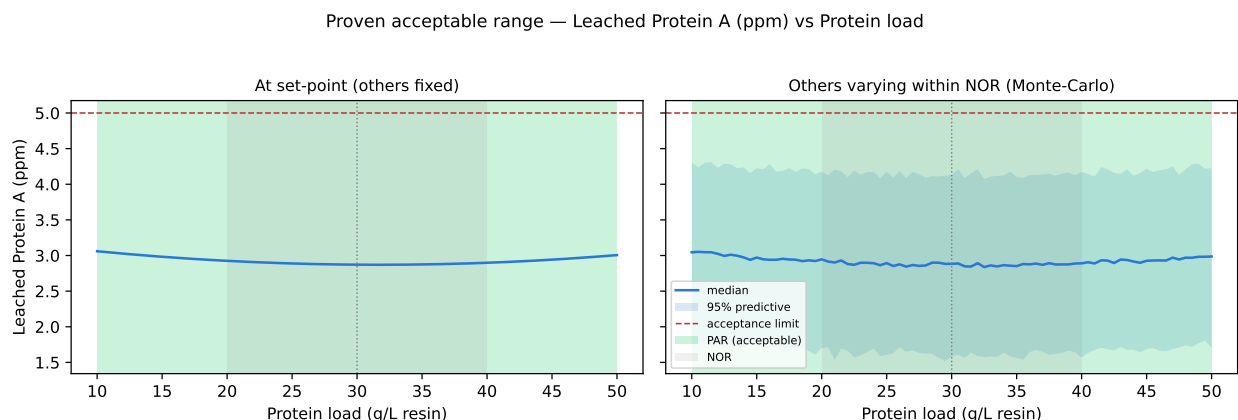


Figure 7.1: Proven acceptable range for leached Protein A against its governing parameter, at set-point (left) and with the other parameters varying within their NOR (right). The acceptable region is shaded green and the acceptance limit is drawn.

For pool HCP no proven acceptable range is returned for any parameter under either analysis, and the reason is that the capture pool does not meet the drug substance limit at any setting studied. At the set-point the model predicts 7,216 ng/mg against a criterion of 100 ng/mg, and the lowest value anywhere in the region is 1,531 ng/mg. Figure 7.2 shows this directly: the acceptance limit lies below the plotted response across the whole range of elution buffer pH, so no region is shaded.

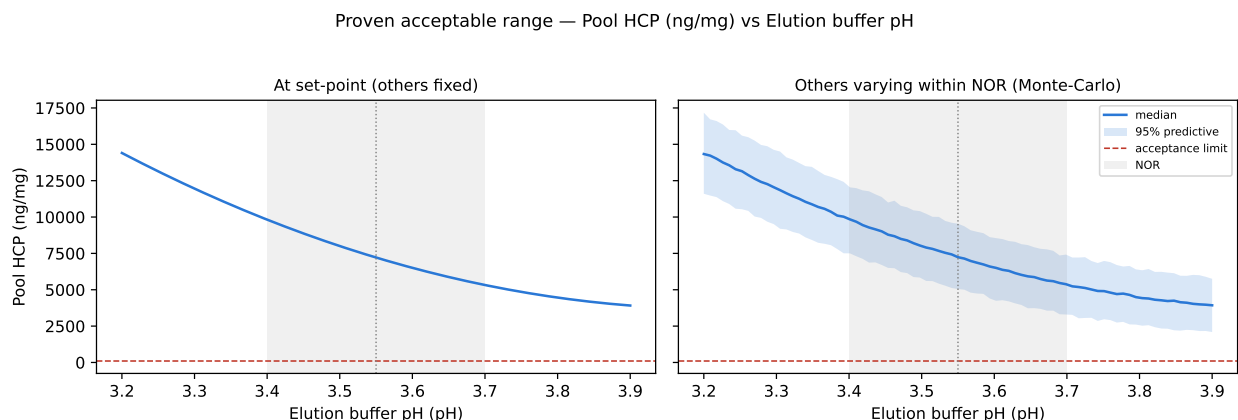


Figure 7.2: Pool HCP against its governing parameter, at set-point (left) and with the other parameters varying within their NOR (right). The drug substance acceptance limit lies below the response everywhere, so no acceptable region is shaded.

This is the expected result for a capture step and not an adverse finding about it. Host cell protein is cleared by three chromatography steps in series, and the criterion applies to the drug substance and not to an intermediate pool. Since the cation and anion exchange steps reduce host cell protein to acceptable and consistent levels, the acceptable pool HCP leaving capture is linked to the operating conditions of those steps and cannot be fixed here in isolation. The linkage is made in PCR-007 and PCR-008 and consolidated in PCMR-001, and the drug substance outcome is reported in Section 8.

The relationship between this section and Section 6 should be read one way only. The proven acceptable ranges here are univariate statements, one parameter at a time, and the NOR-propagated column is a robustness statement about that same univariate range. The design space is the multivariate region. Where a whole characterization range is proven acceptable, as it is for three of the four parameters, the binding constraint on the operating region is not that univariate range at all; it is the corner of the multivariate region, which is why Section 6 names the corners. Every range in Table 7.1 is bounded by the characterization ranges and by the fitted model, and none of them extends beyond the region the study covered.

8 Process capability and robustness

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every process parameter varying inside its normal operating range. Table 8.1 gives the result for the three attributes this step governs, as a one-sided Cpk against the drug substance criterion.

Table 8.1: Commercial-scale capability at the drug substance for the attributes the Protein A step governs.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 $\pm$ 4.6	6.1
Residual DNA	VL	upper	0-0.001	0.0001 $\pm$ 0	10.8
Leached Protein A	L-M	upper	0-5	0.003 $\pm$ 0.0006	2,785.3

Host cell protein has the tightest capability of the three, at Cpk = 6.14, with a simulated mean of 15.2 ng/mg against a limit of 100 ng/mg, so the margin to the limit is 18 standard deviations of the batch-to-batch distribution. That capability is a property of the three clearance steps acting in series and not of capture alone. This step's contribution is the principal clearance of host cell protein from clarified harvest, and the polishing steps are reported in PCR-007 and PCR-008.

Leached Protein A and residual DNA are far from their criteria at the drug substance, at Cpk = 2,785 and 10.8 respectively. A capability index of that size is not a useful discriminator and should not be read as one. The informative statement for leached Protein A is the level in the capture pool itself, before any downstream clearance is credited: 2.87 ppm at the set-point and no more than 3.16 ppm anywhere inside the normal operating ranges, against a criterion of 5 ppm. The attribute the step sets is in specification at the point it is created.

Step yield is a process performance measure and carries no capability index (there is no acceptance criterion of that kind to compute one against). Inside the normal operating ranges the fitted model predicts a yield of at least 90.8%, against 94.7% at the set-point. The nominal batch recovered 97.2% across this step.

Two bounds apply to every figure in this section. All of them are estimated from qualified scale-down models with parameters varying inside their NORs, so they describe the process as it is intended to be run and not its behaviour at the edges of the design space. All of them are predictions to be confirmed at commercial scale during Stage 2. The lowest capability anywhere in the drug substance process is Cpk

= 1.51, which belongs to a viral clearance attribute set at another step and is reported in PCMR-001.

9 Parameter classification

Each parameter was classified under SOP-4001 from its demonstrated effect and from the risk that it leaves the design space in routine operation. The resulting classes are in Table 4.1, and the reasoning for each is below.

- Protein load is a well-controlled critical process parameter. It has a significant effect on pool HCP and interacts with elution buffer pH, so it is linked to a quality attribute. It is set by mass balance from a measured load concentration and column volume, which makes an excursion outside the design space unlikely, so it is classified as well controlled and not as critical.
- Elution buffer pH is a well-controlled critical process parameter. It carries the largest effect on pool HCP of any parameter studied, it is the source of the only curvature that reaches significance, and it also conditions the feed to the low-pH inactivation step. It is nonetheless easy to control: each buffer lot is released on measured pH and the elution is monitored in line. The deviation described in Section 13 shows why that release test matters.
- Load flow rate is a key process parameter. Its effect on pool HCP does not reach significance in the predictive model ($p = 0.060$ in Table 5.7), which is marginal and is not treated as a demonstrated effect, and it has no significant effect on leached Protein A (Table 14.9). Its effect on step yield is significant both alone and through its interaction with protein load (Table 14.8). It is controlled for process consistency and not for product quality.
- End of pool collect is a key process parameter. It carries the largest effect on step yield of any parameter in the study, and it also has a significant effect on pool HCP, at 20.1% of the elution pH coefficient in magnitude, and pool HCP is controlled to its drug substance criterion by the downstream polishing steps and not here. It is therefore classified on process consistency.
- Operating temperature is a general process parameter. The univariate assessment linked it to no quality attribute across its characterization range, and no mechanism is known by which it could affect the affinity interaction over that range.
- Bed height is a general process parameter. It was assessed univariately, it is linked to no quality attribute, and its effect on the separation acts through residence time, which is already characterized through protein load and load flow rate.

Two of these classifications rest on a null result, and both null results are kept rather than dropped. Operating temperature and bed height stay in the knowledge space with their characterization ranges attached, as evidence that the step is robust to them over those ranges. The same applies to load flow rate and end of pool collect with respect to product quality. Neither acts on the attribute this step sets, and the effect end of pool collect carries on pool HCP is small and is managed by the downstream

polishing steps, so both are classified on process consistency and their ranges are justified on that basis.

The summary is 2 well-controlled critical process parameters, 2 key process parameters and 2 general process parameters, and 0 critical process parameters. No parameter at this step requires the most stringent class. That outcome matches the platform history. Affinity capture is an inherently robust separation, and the two quality-linked parameters are both measured directly before use.

10 Contribution to the control strategy

The capture step contributes five things to the drug substance control strategy.

First, it controls the two quality-linked parameters to the normal operating ranges in Table 4.1: protein load to 20–40 g/L resin and elution buffer pH to 3.4–3.7. Both ranges sit inside the design space of Section 6 with margin, and both parameters are verified before use rather than inferred afterwards.

Second, it controls the two key process parameters, load flow rate and end of pool collect, to their normal operating ranges for process consistency, since neither is linked to a quality attribute and both are included because holding them makes the step's yield and pool volume predictable for the operations that follow.

Third, it carries in-process controls on the elution pool. Pool host cell protein and leached Protein A are measured on every batch (AMV-3012 and AMV-3016), and step yield is closed by mass balance. These are monitoring points and not acceptance limits at this step, and they are the input the downstream clearance credit is taken against.

Fourth, it carries life-cycle controls on the resin. Packing, sanitization and cycle accounting run under SOP-2008, and the characterization reported here was executed on resin early in its validated life (cycle number was held constant across the study). A change of resin is outside this control strategy and would require independent characterization under RA-004.

Fifth, it carries a material control that follows directly from Section 13. Each elution buffer lot is released on measured pH before use, because elution buffer pH is the parameter with the largest effect on pool HCP and a preparation error is not otherwise detectable until the pool is assayed.

The step does not control host cell protein to its drug substance criterion, and the control strategy does not claim that it does. The nominal batch leaves capture at 9,125 ng/mg. Cation exchange then reduces host cell protein by a factor of 78 and anion exchange by a further factor of 6.2, giving 18.9 ng/mg at the drug substance for that batch. The mean across the 2,000 simulated batches of Table 8.1 is 15.2 ng/mg, since the simulation varies every parameter inside its normal operating range and the nominal batch is a single point in that distribution. Those credits belong to PCR-007 and PCR-008. The step likewise makes no viral clearance claim. Clearance across affinity capture is well documented for this class of process (CMC Biotech Working Group 2009), but the modular claim for A-Mab rests on low-pH inactivation, anion exchange and small-virus retentive filtration, and it is consolidated in PCMR-001 (International Council for Harmonisation 2023a).

11 Discussion

The capture step is well understood over the ranges studied, and the understanding is concentrated in one relationship. Pool host cell protein is governed by elution buffer pH and protein load together, with a significant interaction and significant curvature in pH, and the response-surface model describes that relationship well enough to predict a held-out run. Everything else the study found is either a process performance effect or a null result. That is a simple outcome for a step of this importance, and it is consistent with the platform history summarized in Section 2.

The result that most constrains what can be claimed is the leached Protein A model. The attribute this step sets is the one for which no factor effect could be resolved, and the reason is visible in the centre points: pure error of 33.8 % leaves little room for a real effect to emerge. The conclusion is not that the operating parameters cannot influence ligand release. It is that they do not influence it enough to be seen against the step's own run-to-run variation across the ranges studied, which is the more useful statement for a control strategy and the weaker one for a mechanistic claim.

Confidence in the scale-down model rests on the qualification described in Section 3 and on the agreement between the fitted effects and the platform expectation. The agreement is closer on the effects than on the absolute level. The centre-point mean for pool HCP in the response-surface design is 7,564 ng/mg, against 9,125 ng/mg for the nominal batch at commercial equivalent conditions. That difference is small beside the 16-fold span the model covers across the characterized region, and it is larger than the scatter of the centre points themselves (standard deviation 357 ng/mg on 4 runs). It is quoted here as the limit on how precisely the absolute level transfers to the commercial step. The effects are what the design space rests on.

The effects the platform expected are the effects the study found. Pool HCP rises with protein load, rises as elution buffer pH falls, and the two interact. Step yield falls when a high load is combined with a high linear velocity. Two significant findings were not anticipated in Section 2. The curvature in elution buffer pH was not predicted, and the effect of end of pool collect on pool HCP has the opposite sign to the expectation that a later cut trades purity for yield. Both are explained mechanistically in Section 5, and neither changes a classification or a range.

Three limitations bound the conclusions. The design space is not confirmed at scale at the edges of its ranges, since the study was executed on a scale-down model and the commercial step is run near its set-point. Resin cycle number was held constant, so the study says nothing about performance late in the resin's validated life, and that remains under the life-cycle controls of SOP-2008. The study used a single representative clarified harvest, so it characterizes this step's parameters and not its sensitivity to a feed that differs from the platform expectation (the harvest step that produces that feed is characterized in PCR-004).

2 deviations were recorded, described in full in Section 13. One was an elution buffer prepared below its target pH, which affected a single response-surface run. The other was an out-of-trend leached Protein A result that was re-assayed. Both are departures from the approved protocol, and neither changed a fitted effect, a classification or the operating region. No parameter at this step needed the most stringent class of control, and the two quality-linked parameters are both well controlled by measurement before use.

12 Conclusions

The Protein A capture step is characterized over the ranges in Table 4.1 and its operating region is defined. 6 parameters were assessed, 4 in a multivariate design and 2 univariately, giving 2 well-controlled critical process parameters, 2 key process parameters and 2 general process parameters. The step has no critical process parameter.

Leached Protein A, the attribute this step sets, met its acceptance criterion of 0-5 ppm in both designs, at a highest measured value of 4.61 ppm, and the fitted model resolves no effect of any studied parameter on it. The design space caps elution buffer pH at 3.86, because above that point the upper bound of the 95 % predictive interval for this attribute reaches the criterion (Section 7). Pool host cell protein is governed by elution buffer pH and protein load through a response surface that predicts a held-out run with a predicted R^2 of 0.932. At the drug substance (Table 8.1), the tightest of the three attributes this step governs is host cell protein, at $Cpk = 6.14$ with a mean of 15.2 ng/mg against a limit of 100 ng/mg.

2 deviations were investigated and dispositioned as retained, with no effect on the classifications or on the region defined in Section 6. Every conclusion above is bounded by the characterization ranges, by the fitted models and by the qualified scale-down model they were fitted on, and none of them is confirmed at commercial scale until Stage 2. The step does not by itself assure the drug substance limit for host cell protein and makes no viral clearance claim. These results roll up into PCMR-001.

13 Deviations from the plan

2 deviations from PCP-005 were recorded during execution. Each was investigated, assigned a root cause, impact-assessed against the study conclusions and dispositioned. Both were dispositioned as retained, which means the affected data were kept in the reported analysis. Table 13.1 is the register.

Table 13.1: Deviations recorded during execution of PCP-005.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-005-01	Elution-buffer pH prepared below target on an RSM run	buffer release testing	retained
DEV-005-02	Leached Protein A result initially out-of-trend; re-assayed	analytical review	retained

The two entries differ in kind. The first is a preparation error that put a studied parameter away from its target on one run (response-surface run 23). The second is an analytical result that was rejected on review and repeated. Neither invalidated a design, so no part of this study was re-executed.

13.1 DEV-005-01 — elution buffer prepared below target pH

The elution buffer used on response-surface run 23 was released at pH 3.38 against a target of 3.55, a departure of 0.17 pH units. The deviation was detected during buffer release testing. Elution buffer pH carries the largest effect on pool HCP of any parameter studied (Table 14.5), so a departure of this size is material and is assessed on that basis and not dismissed as small.

The investigation traced the cause to a titration endpoint error during preparation of lot LOT-BUF-5290 (Fern Scientific), which is the buffer lot the deviation record names. The pH was read before the buffer had equilibrated, and the correction added on that reading overshot. No other lot from the same preparation session was used in the study, and the release procedure in SOP-2103 was reinforced with an equilibration hold before the endpoint reading.

The impact assessment turned on three things. The measured pH of 3.38 lies inside the characterization range for elution buffer pH, at a coded position of -0.49, so the affected run remains inside the region the response-surface model covers. Run 23 is an axial run for end of pool collect, so its design setting for elution buffer pH was

the set-point, and at the recorded pH the fitted model predicts 11,961 ng/mg of pool HCP against 8,601 ng/mg at target. The measured result for that run was 8,167 ng/mg, which is close to the prediction at target and gives a standardized residual of -0.49 against the fitted model. The run's influence on the fit is correspondingly small: its Cook's distance is 0.031, against a maximum of 0.424 across the 28 runs of the design.

The run was therefore retained with its design setting. The recorded response is consistent with that setting and not with the offset the buffer record implies, and the leverage the run has on the fit is small, so keeping it does not materially change the fitted surface. The deviation is nonetheless the reason the control strategy in Section 10 keeps buffer pH release testing as a named control.

13.2 DEV-005-02 — leached Protein A result out of trend on first assay

A leached Protein A result was flagged as out of trend during analytical review, before the data set was released for analysis, because the initial value was inconsistent with the other results from the same assay plate and with the rest of the leached Protein A results in the study.

The investigation suspected carryover on the ELISA plate. The retained sample was re-assayed under AMV-3016 on a fresh plate with additional wash steps, and the repeat gave 1.6 ppm (above the method's limit of quantitation of 0.25 ppm). The initial result was rejected under the method's own repeat criteria and the repeat was accepted in its place. The register does not link the deviation to a numbered run, so the affected sample is not identified in the design matrices of Appendix A and Appendix B.

The impact on the study is bounded by two figures. The retest value of 1.6 ppm is well inside the acceptance criterion of 0–5 ppm, and it sits inside the spread the centre points show for this response, which is 33.8 % at the set-point. Since no factor effect on leached Protein A was resolved by either design (Section 5.3), no conclusion of this report depends on the value of any single leached Protein A result. The deviation was dispositioned as retained and no re-execution was required.

14 Appendices

The design matrices, the full effect and coefficient tables and the analytical method summary are given below. They carry the detail the body sections draw on.

14.1 Appendix A — Screening design matrix and responses

The coded settings of the screening design are given in Table 14.1 and the measured responses in Table 14.2. Runs are listed in design order, not in execution order; execution was randomized.

Table 14.1: Screening design, coded settings. Factor letters are given in Table 4.2.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 14.2: Screening design, measured responses.

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	factorial	9,965.7	0.918	2.64

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
2	factorial	9,515.4	0.99	3
3	factorial	9,022.6	0.928	3.03
4	factorial	7,893.7	0.98	2.98
5	factorial	3,211.7	0.915	2.88
6	factorial	2,523.1	0.986	3.82
7	factorial	2,540.2	0.929	3.3
8	factorial	3,108.1	0.96	2.43
9	factorial	16,568.2	0.913	2.47
10	factorial	19,737.9	0.99	2.66
11	factorial	18,271.3	0.857	2.67
12	factorial	21,244.6	0.936	3.54
13	factorial	6,791.9	0.911	2.77
14	factorial	7,875.0	0.976	2.29
15	factorial	7,655.0	0.891	2.88
16	factorial	7,791.9	0.927	3.14
17	center	6,436.2	0.931	2.91
18	center	9,652.2	0.933	3.07
19	center	8,221.9	0.942	3.86

14.2 Appendix B — Response-surface design matrix and responses

The coded settings of the face-centred central composite design are given in Table 14.3 and the measured responses in Table 14.4.

Table 14.3: Response-surface design, coded settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1

Run	Type	A	B	C	D
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 14.4: Response-surface design, measured responses.

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	factorial	10,011.3	0.93	3.42
2	factorial	9,269.0	0.99	3.06
3	factorial	10,650.9	0.928	2.53
4	factorial	7,715.2	0.988	2.48
5	factorial	2,992.5	0.923	3.27
6	factorial	2,149.1	0.977	1.89
7	factorial	2,643.4	0.908	4.61
8	factorial	2,663.4	0.99	3.13
9	factorial	25,114.3	0.89	3.38
10	factorial	20,388.5	0.99	2.49
11	factorial	22,755.7	0.861	2.12
12	factorial	16,694.1	0.924	3.03
13	factorial	7,882.2	0.895	3.63
14	factorial	6,577.7	0.97	2.72
15	factorial	6,798.5	0.844	3.06
16	factorial	6,100.0	0.962	2.82
17	axial	3,900.5	0.935	2.71
18	axial	9,377.3	0.935	3.37
19	axial	13,731.8	0.966	3.77
20	axial	4,121.9	0.955	3.8
21	axial	7,571.5	0.955	2.3
22	axial	8,097.7	0.955	2.54
23	axial	8,167.0	0.908	2.07
24	axial	6,460.6	0.953	2.69
25	center	8,036.2	0.94	2.75
26	center	7,616.5	0.967	1.7
27	center	7,206.4	0.924	4.06

Run	Type	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
28	center	7,396.1	0.954	2.91

14.3 Appendix C — Full effect and coefficient tables

The complete screening effect tables for all three responses are given in Table 14.5 to Table 14.7. Effects are twice the coded coefficient, so each is the change in the response across the full characterization range of that term.

Table 14.5: Screening effects on pool HCP.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	
BD	-433.1	-216.5	350	-0.619	0.553	
C	167.3	83.7	350	0.239	0.817	
CD	-70.6	-35.3	350	-0.101	0.922	
BC	6.1	3.0	350	0.00868	0.993	

Table 14.6: Screening effects on step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*
AB	0.00443	0.00221	0.00208	1.06	0.319	
AD	0.0037	0.00185	0.00208	0.889	0.4	
BC	0.0035	0.00175	0.00208	0.84	0.425	
B	-0.00236	-0.00118	0.00208	-0.568	0.586	

Table 14.7: Screening effects on leached Protein A. No term is significant.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.329	0.165	0.128	1.28	0.235	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.208	-0.104	0.128	-0.811	0.441	
BD	-0.189	-0.0944	0.128	-0.736	0.482	
BC	-0.182	-0.0912	0.128	-0.712	0.497	
C	0.18	0.0898	0.128	0.701	0.503	
D	0.152	0.0758	0.128	0.591	0.571	
AB	-0.133	-0.0664	0.128	-0.518	0.618	
CD	-0.0987	-0.0494	0.128	-0.385	0.71	
B	0.0652	0.0326	0.128	0.254	0.806	
AD	0.0567	0.0284	0.128	0.221	0.83	

The response-surface coefficient tables for the two responses not shown in Section 5.3 are given in Table 14.8 and Table 14.9, with their ANOVA partitions in Table 14.10 and Table 14.11.

Table 14.8: Response-surface coefficients for step yield, in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.947	0.00554	171	3.5e-23	***
A	-0.0165	0.00377	-4.36	0.000767	***
B	-0.00243	0.00377	-0.643	0.531	
C	-0.00891	0.00377	-2.36	0.0346	*
D	0.0365	0.00377	9.68	2.62e-07	***
AB	0.00266	0.004	0.665	0.517	
AC	-0.00914	0.004	-2.28	0.0399	*
AD	0.00629	0.004	1.57	0.14	
BC	0.00243	0.004	0.606	0.555	
BD	0.00291	0.004	0.727	0.48	
CD	0.00219	0.004	0.547	0.594	
A ²	-0.0128	0.00997	-1.29	0.221	
B ²	0.012	0.00997	1.2	0.252	
C ²	0.00681	0.00997	0.683	0.507	
D ²	-0.0178	0.00997	-1.78	0.0976	

Table 14.9: Response-surface coefficients for leached Protein A, in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.87	0.22	13	7.67e-09	***
A	-0.027	0.15	-0.18	0.86	
B	0.148	0.15	0.989	0.341	
C	0.00918	0.15	0.0612	0.952	
D	-0.21	0.15	-1.4	0.185	
AB	-0.0127	0.159	-0.0796	0.938	
AC	-0.143	0.159	-0.899	0.385	

Term	Coef.	Std. err.	t	p-value	Sig.
AD	0.132	0.159	0.831	0.421	
BC	0.269	0.159	1.69	0.115	
BD	-0.226	0.159	-1.42	0.179	
CD	0.167	0.159	1.05	0.314	
A ²	0.162	0.397	0.408	0.69	
B ²	0.906	0.397	2.29	0.0397	*
C ²	-0.458	0.397	-1.15	0.269	
D ²	-0.502	0.397	-1.27	0.228	

Table 14.10: ANOVA partition for the step yield model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.0348	14	0.00249	9.69	0.000105
Residual	0.00333	13	0.000257	nan	nan
Lack of fit	0.00229	10	0.000229	0.659	0.731
Pure error	0.00104	3	0.000348	nan	nan

Table 14.11: ANOVA partition for the leached Protein A model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	6.9	14	0.493	1.22	0.365
Residual	5.27	13	0.406	nan	nan
Lack of fit	2.47	10	0.247	0.264	0.953
Pure error	2.8	3	0.935	nan	nan

14.4 Appendix D — Analytical methods summary

The validated performance of the two ELISA methods used at this step is given in Table 14.12. The full list of methods supporting the step, including those used for the in-process panel, is in Table 1.2.

Table 14.12: Validated performance of the two ELISA methods used for the reported impurity responses.

Method	Title	Precision (%RSD)	LOQ	Accuracy (%)
AMV-3016	Leached Protein A by ELISA	6.5	0.25 ppm	97.4
AMV-3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95

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